

Øving 11

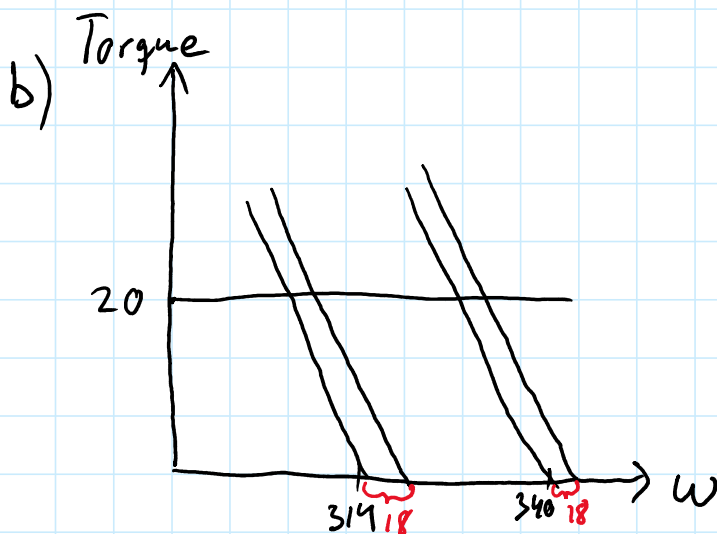
torsdag 27. mars 2025

12:07

Problem 1

$$a) \omega_s = \frac{2\pi f}{\frac{P}{2}} = 2\pi f = 100\pi$$

$$s = \frac{\omega_s - \omega_m}{\omega_s} = \frac{100\pi - 295,31}{100\pi} = \frac{18,85}{100\pi} = 0,06$$



$$s_2 = \frac{18,85}{340} = 0,055$$

$$P = (340 - 18,85) \cdot 20 \text{ W} = 6423 \text{ W}$$

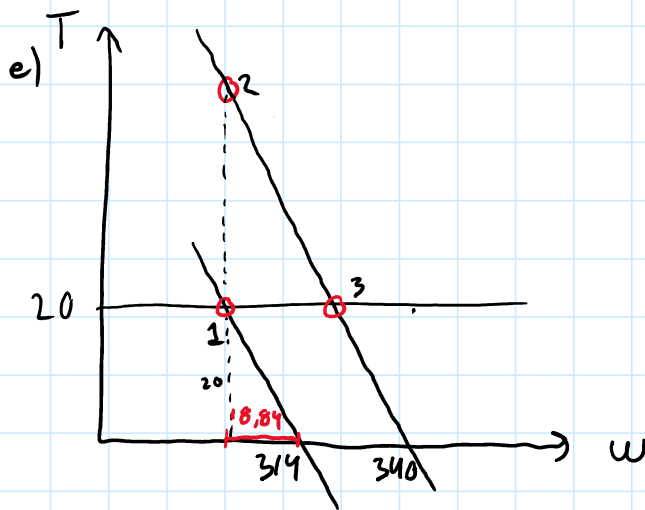
$$c) \frac{V_{ac,1}}{f_{ac,1}} = \frac{V_{ac,2}}{f_{ac,2}} \Rightarrow V_{ac,2} = \frac{V_{ac,1} \cdot f_{ac,2}}{f_{ac,1}}$$
$$= \frac{200 \cdot \frac{340}{2\pi}}{50} \text{ V} = 216,45 \text{ V}$$

$$m_{a,1} = \frac{V_{ac,1} \cdot \sqrt{2}}{V_{dc}} = \frac{200 \cdot \sqrt{2}}{400} = 0,707$$

$$m_{a,2} = \frac{216,45 \cdot \sqrt{2}}{400} = 0,767$$

$$d) T \cdot \omega_m = I_{dc} \cdot V_{dc} \Rightarrow I_{dc,1} = 400 = 14,765 A$$

$$I_{dc,2} = \frac{P}{V_{dc}} = \frac{6423}{400} = 16,06 A$$



Go from 1 to 2 to 3

$$f) T_{o,1} \left(1 - \frac{295,31}{314,16} \right) = 20$$

$$T_{o,2} = 333,3 N_m$$

$$T_{o,2} = 333,3 - \frac{340}{314,16} N_m = 360,4 N_m$$

$$T_{new} = 360,4 \left(1 - \frac{295,31}{340} \right) = 47,4 N_m$$

$$T_{net} = 47,4 - 20 N_m = 27,4 N_m$$

$$\frac{d\omega_m}{dt} = \frac{27,4}{0,1} = 274 \text{ rad/s}^2$$

$$g) T_m(\omega_m) = T_o \left(1 - \frac{\omega_m}{\omega_{ac}} \right)$$

$$T_m(\omega_m) - T_{load} = J \frac{d\omega_m}{dt}$$

$$360,4 \left(1 - \frac{\omega_m}{340} \right) - 20 = 0,1 \frac{d\omega_m}{dt}$$

$$\frac{d\omega_m}{dt} = 3404 - 10,6 \omega_m$$

$$\frac{dw_m}{dt} = 3404 - 10,6 w_m$$

$$\int \frac{dw_m}{3404 - 10,6 w_m} = \int dt$$

$$u = 3404 - 10,6 w_m$$

$$\Rightarrow \frac{du}{dw_m} = -10,6 \Rightarrow dw_m = -\frac{1}{10,6} du$$

$$\int \frac{1}{u} \left(-\frac{1}{10,6}\right) du = -\frac{1}{10,6} \ln|3404 - 10,6 w_m|$$

$$-\frac{1}{10,6} \ln|3404 - 10,6 w_m| = t + C$$

$$\Rightarrow 3404 - 10,6 w_m = e^{-10,6(t+C)} = A e^{-10,6t}$$

$$w_m(t) = \frac{3404 - A e^{-10,6t}}{10,6}$$

$$w_m(0) = 295,31$$

$$295,31 = \frac{3404 - A}{10,6} \Rightarrow 273,71$$

$$w_m(t) = 321,16 - 25,85 e^{-10,6t}$$

$$\tau = \frac{1}{10,6} = 0,094 \text{ s}$$

Problem 2

$$a) m_a = \frac{2\sqrt{2} V_{LL}}{\sqrt{3} V_{Lc}} = \frac{2\sqrt{2} \cdot 470}{\sqrt{3} \cdot 1000} = 0,767$$

$$b) \omega_m = 2\pi f_s (1-s)$$

$$P = T_{wm} = 752097$$

$$\varphi = 0,451, \quad S = \frac{3}{\sqrt{3}} \cdot V_{LL} \cdot I_{LL}$$

$$\operatorname{Re}(\sqrt{3} V_{LL} I_{LL} \angle -0,451) = 752097$$

$$\sqrt{3} I \operatorname{Re}(0,9 - 0,436j) = 1600,2$$

$$I = \frac{1600,2}{\sqrt{3} \cdot 0,9} = 1026,4 \text{ A}$$

Problem 3

a) Load torque:

$$V = \frac{100 \text{ km}}{h} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot \frac{1 \text{ h}}{3600 \text{ s}} = 27,78 \text{ m/s}$$

$$F_{\text{Load}} = 300 + 0,5 v^2$$

$$= 300 + 0,5 \cdot (27,78)^2 = 685,9 \text{ N}$$

$$T_{\text{Load}} = F_{\text{Load}} \cdot r = 685,9 \cdot 0,3 = 205,7 \text{ Nm}$$

$$\omega_{\text{mec}} = \frac{v}{r} = \frac{27,78}{0,3} \text{ rad/s} = 92,6 \text{ rad/s}$$

$$\omega_{mec} = \frac{2\pi f_{el}}{p}$$

$$f_{el} = \frac{p \omega_{mec}}{2\pi} = \frac{3 \cdot 92,6}{2\pi} = 44,23 \text{ Hz}$$

$$P_{mech} = T_{load} \cdot \omega_{mec} = 206 \cdot 92,6 = 19 \text{ kW}$$

$$t = \frac{70 \text{ kWh}}{19 \text{ kW}} = 3,7 \text{ h}$$

$$b) v = 50 \text{ km/h} = 13,89 \text{ m/s}$$

$$F_{load} = 300 + 0,5 v^2 = 300 + 0,5 \cdot (13,89)^2 = 396,5 \text{ N}$$

$$T_{load} = F_{load} r = 396,5 \cdot 0,3 = 119 \text{ Nm}$$

$$\omega_{mec} = \frac{13,89}{0,3} = 46,3 \text{ rad/s}$$

$$P_{mech} = T_{load} \omega_{mec} = 119 \cdot 46,3 \text{ W} = 5,5 \text{ kW}$$

$$t = \frac{70 \text{ kWh}}{5,5 \text{ kW}} = 12,7 \text{ h}$$

$$c) 12,7 \text{ h} \cdot 50 \text{ km/h} = 635,6 \text{ km}$$

$$3,7 \text{ h} \cdot 100 \text{ km/h} = 370 \text{ km}$$

$$3,7h \cdot 100 \text{ km/h} = 370 \text{ km}$$

$$d) E_k = \frac{1}{2} m v_0^2 = \frac{1}{2} \cdot 2000 \cdot (16,67)^2 = 278889 \text{ J}$$

$$t = \frac{E_k}{P} = \frac{278889}{150000} = 1,86 \text{ s}$$

$$v_{\text{avg}} = \frac{v_0 + 0}{2} = \frac{16,67}{2} = 8,33 \text{ m/s}$$

$$d = v_{\text{avg}} \cdot t = 8,33 \cdot 1,86 = 15,5 \text{ m}$$

$$e) E_k = \frac{1}{2} m v^2 = \frac{1}{2} \cdot 2000 \cdot (27,78)^2 = 771600 \text{ J}$$

$$t = \frac{E_k}{P} = \frac{771600 \text{ J}}{300 \text{ kW}} = 2,57 \text{ s}$$

$$I_{\text{dc}} \cdot 600 \text{ V} = 300 \text{ kW}$$

$$I_{\text{dc}} = 500 \text{ A}$$

$$f) \sqrt{3} I_m V_{\text{LL}} = 300 \text{ kW}$$

$$I_m = 433 \text{ A}$$